Modern Ciphers

1. What are Modern Ciphers?

- Encryption algorithms designed for use with computers and modern technology.
- Two main types:
    1. **Stream Ciphers**
    2. **Block Ciphers**

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2. Stream Ciphers

- **What is a Stream Cipher?**
    - Encrypts data one bit or byte at a time.
    - Combines plaintext with a pseudorandom key stream (generated using a key).
    - Example: **Vernam Cipher (One-Time Pad)**.
- **Advantages:**
    - Fast and efficient for real-time communication (e.g., wireless networks).
- **Disadvantages:**
    - Vulnerable if the same key stream is reused.

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3. Block Ciphers

- **What is a Block Cipher?**
    - Encrypts data in fixed-size blocks (e.g., 64 bits or 128 bits).
    - Processes one block at a time using a secret key.
    - Example: **Data Encryption Standard (DES)**.
- **Advantages:**
    - More secure than stream ciphers for large data.
    - Widely used in applications like file encryption and secure communication.
- **Disadvantages:**

- Slower than stream ciphers for real-time communication.

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4. Data Encryption Standard (DES)

https://www.youtube.com/watch?v=j53iXhTSi_s

- **What is DES?**
  - A symmetric-key block cipher developed by IBM in the 1970s.
  - Uses a 56-bit key to encrypt 64-bit blocks of data.
  - Widely used in the past but now considered insecure due to its short key length.
- **How DES Works:**
  1. **Initial Permutation (IP):** Rearranges the bits of the plaintext block.
  2. **16 Rounds of Feistel Network:** Each round uses a subkey and performs substitution and permutation.
  3. **Final Permutation (FP):** Produces the ciphertext block.
- **Weaknesses:**
  - 56-bit key is too short (vulnerable to brute-force attacks).
  - Replaced by **AES (Advanced Encryption Standard)**.

How it Works (Simplified):

1. **Split:** Divide a data block (e.g., 64 bits) into two equal halves: Left (L) and Right (R).
2. **Round Function (F):** The Right half (R) is processed by a function (F) that uses a part of the secret key (sub-key).
3. **XOR:** The output of F is XORed (exclusive OR) with the Left half (L).
4. **Swap:** The original Right half (R) becomes the new Left half, and the XORed result becomes the new Right half (L' = R, R' = L XOR F(R, Key)).
5. **Repeat:** This process iterates for several rounds (e.g., 16 for DES).
6. **Decrypt:** Decryption uses the *exact* same steps but applies the sub-keys in reverse

order, leveraging XOR's property (A XOR B XOR B = A) to undo the encryption.

https://medium.com/@kosisochukwuchibuike5/how-des-works-a-beginners-guide-to-the-encryption-algorithm-that-secures-your-data-20f9a357508f

Example (DES)

Our Consistent Rules:

- *Block size: 4 bits*
- *Key size: 6 bits*
- *Rounds: 2*
- *Expansion: 2 bits → 4 bits (duplicate each bit)*
- *S-Box: 4×4 table (2 bits in → 1 bit out)*
- *Permutation: Simple fixed rule*

New S-Box (2×2):

```text
Input (2 bits) | Output (1 bit)
00             | 1
01             | 0
10             | 1
11             | 0
```

Plaintext: `1011`

Key: `110010`

Step 1: IP (swap halves)

`1011` → `10 11` → Swap → `1110`

Step 2: Split

L0 = `11`, R0 = `10`

Round 1:

K1 = `1100`

f(R0, K1):

1. Expand R0: `10` → `1100`
2. XOR: `1100 ⊕ 1100` = `0000`
3. Split: `00` and `00`
4. S-Box: `00` → `1`, `00` → `1`
5. Combine: `11`
6. Permutation (swap bits): `11` → `11` (same!)

f(R0, K1) = `11`

Update:
L1 = R0 = `10`
R1 = L0 ⊕ f(R0, K1) = `11 ⊕ 11` = `00`

After Round 1: L1=`10`, R1=`00`

Round 2:

K2 = `0010`

f(R1, K2):

1. Expand R1: `00` → `0000`
2. XOR: `0000 ⊕ 0010` = `0010`
3. Split: `00` and `10`
4. S-Box: `00` → `1`, `10` → `1`
5. Combine: `11`
6. Permutation: `11` → `11`

f(R1, K2) = `11`

Update:
L2 = R1 = `00`
R2 = L1 ⊕ f(R1, K2) = `10 ⊕ 11` = `01`

After Round 2: L2=`00`, R2=`01`

Step 3: Combine and Final Permutation

Combine: L2 + R2 = `00 01` = `0001`
Final IP^{-1} (swap halves): `0001` → `00 01` → Swap → `0100`

Final Ciphertext: `0100`

Verification by Decryption:

Ciphertext: `0100`
Key: `110010`

Step 1: IP on ciphertext
`0100` → `01 00` → Swap → `0001`

Step 2: Split
L2= `00` , R2= `01`

Step 3: Round 2 Reverse (use K2)
Compute f(L2, K2):

1. Expand L2: `00` → `0000`
2. XOR: `0000 ⊕ 0010` = `0010`
3. Split: `00` and `10`
4. S-Box: `00` → `1` , `10` → `1`
5. Combine: `11`
6. Permute: `11` → `11`

f(L2, K2) = `11`

L1 = R2 ⊕ f(L2, K2) = `01 ⊕ 11` = `10` ✓
R1 = L2 = `00` ✓

Step 4: Round 1 Reverse (use K1)
Compute f(L1, K1):

1. Expand L1: `10` → `1100`
2. XOR: `1100 ⊕ 1100` = `0000`
3. Split: `00` and `00`
4. S-Box: `00` → `1` , `00` → `1`
5. Combine: `11`
6. Permute: `11` → `11`

f(L1, K1) = `11`

L0 = R1 ⊕ f(L1, K1) = `00 ⊕ 11` = `11` ✓

R0 = L1 = `10` ✓

Step 5: Combine and IP^{-1}

Combine: `11 10` = `1110`

IP^{-1}: `11 10` → Swap → `1011` ✓

Original plaintext recovered! `1011`

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5. Double DES

- **What is Double DES?**
  - A variant of DES that applies the DES algorithm twice using two different keys.
  - Encryption: Ciphertext = DES(Key2, DES(Key1, Plaintext)).
  - Decryption: Plaintext = DES^{-1}(Key1, DES^{-1}(Key2, Ciphertext)).
- **Why Use Double DES?**
  - Intended to increase security by doubling the key length (56 + 56 = 112 bits).
- **Weakness:**
  - Vulnerable to **Meet-in-the-Middle Attack**, which reduces effective security to 57 bits.

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6. Triple DES (3DES)

- **What is Triple DES?**
  - Applies the DES algorithm three times using two or three different keys.
  - Two common modes:
    1. **DES-EEE3:** Encrypts with Key1, then Key2, then Key3.
    2. **DES-EDE3:** Encrypts with Key1, decrypts with Key2, then encrypts with Key3.
- **Key Length:**
  - Uses 168-bit keys (3 x 56-bit keys).
- **Advantages:**
  - More secure than DES and Double DES.

- Backward compatible with DES.

- **Disadvantages:**
  - Slower than DES due to triple encryption.
  - Being phased out in favor of AES.

Summary Table

Cipher	Type	Key Feature
Stream Cipher	Modern Cipher	Encrypts data one bit/byte at a time.
Block Cipher	Modern Cipher	Encrypts data in fixed-size blocks (e.g., 64 bits).
DES	Block Cipher	Uses a 56-bit key to encrypt 64-bit blocks.
Double DES	Block Cipher	Applies DES twice with two keys (112-bit key length).
Triple DES (3DES)	Block Cipher	Applies DES three times with two or three keys (168-bit key length).

Key Points to Remember

1. **Stream Ciphers:** Fast and efficient for real-time communication but vulnerable to key reuse.
2. **Block Ciphers:** More secure for large data but slower than stream ciphers.
3. **DES:** An outdated block cipher with a 56-bit key.
4. **Double DES:** Vulnerable to Meet-in-the-Middle Attack.
5. **Triple DES (3DES):** More secure than DES but slower and being replaced by AES.